

The Effect of Relaxation on Mother's Anxiety and Maternal–Fetal Attachment in Primiparous IVF Mothers

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Abstract: Objectives: Infertility is one of the main stressful events of life and a crisis affecting various dimensions of infertile couples' lives. Relaxation, as a strategy, can be effective in the individuals' resistance to stress and anxiety. Therefore, the present study aimed to assess the effect of relaxation on mother's anxiety and maternal–fetal attachment in primiparous women who used In Vitro Fertilization (IVF) to get pregnant.

Methods: In this clinical trial, 80 primiparous women who had used IVF to get pregnant were randomly divided into a control and an intervention group. The intervention group took part in four 90-min relaxation classes in addition to receiving the routine pregnancy care. Anxiety and maternal–fetal attachment were evaluated before and one month after the intervention. Also, maternal–infantile attachment score was computed after delivery. The statistical analyses were performed using the SPSS statistical software. Chi-square and independent t-test was used to analyze the data.

Results: The results revealed no significant difference between the two groups regarding the mean scores of anxiety and attachment before the intervention. However, a significant difference was observed in this regard after the intervention (41.4 ± 4.1 vs. 51.0 ± 10.4 , $P < 0.01$).

Conclusion: The study results showed that relaxation training was effective in reduction of anxiety and increase of maternal–fetal attachment in the women who had used IVF to get pregnant. Although there are other methods for reducing anxiety, relaxing can be used for IVF women due to its ease of use and its wide acceptance.

Keywords: Infertility ■ Relaxation ■ Anxiety ■ Fetal attachment ■ Infantile attachment

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INTRODUCTION

Infertility is one of the bitterest and most stressful events of marital life. According to statistics, 10–15% of couples in reproductive ages suffer from infertility

problems.¹ Stress resulting from infertility increases for the couples who actively seek for treatment of their disorder over time. Uncertainty about success or failure of the treatment increases the anxiety, as well.² Almost 32% of the women are exposed to mental problems at the beginning stages of infertility treatment.³ The women undergoing In Vitro Fertilization (IVF) are often depressed and anxious due to infertility, uncertainty, and low success rate of the treatment methods.⁴ Mental pressure and anxiety, in turn, threaten the treatment outcomes and endanger pregnancy followed by IVF.⁵ In general, anxiety during pregnancy and delivery is accompanied by negative effects. Long-term anxiety, by stimulation of the autonomic nervous system, leads to decrease in uteroplacental blood flow, reduction of oxygenation to uterus, abnormal fetal heart rate, and increase in the probability of preterm delivery. In high-risk pregnancies with the history of infertility,⁶ hormones such as catecholamine, cortisol, epinephrine, and beta-endorphins which are secreted in response to anxiety disturb progression in cervical dilation, affect uterine smooth muscles, reduce uterine contraction capacity, and decrease the efficiency of uterine contractions during delivery. These will eventually lengthen the delivery, increase pain, and create anxiety.⁷

Due to anxiety, depression, and feeling guilty, infertile couples are prone to mental problems. Yet, psychological and educational intervention can reduce depression and anxiety and improve the mental health of the pregnant women with the history of infertility.^{8–11} Such interventions must be designed in a way that they reduce the pregnant women's anxiety and, consequently, increase their survival, attachment, and fertility health. Talking to these women about risks, complications, and success rate of pregnancy through educational methods can improve the pregnancy status.¹² Since having insufficient information about pregnancy can intensify anxiety, increasing the women's knowledge level most often has a positive impact on the pregnancy outcome. Thus, one of the major responsibilities of the healthcare staff is reducing the infertile women's anxiety during treatment and pregnancy, so that treatment measures will result in more

desirable outcomes. In fact, pregnant women's unpreparedness is the main reason for increase of medical interventions complications, and training the women during pregnancy prepares them for having a healthy pregnancy period.^{13,14} In the study conducted by Gamble, consultation reduced the pregnant women's anxiety. In spite of the fact that the intervention group women showed a lower anxiety level in that study, the difference was not statistically significant.¹⁵ Similarly, Ryding et al. indicated that consultation and training the pregnant women had a positive effect on reduction of anxiety.¹⁶ According to the obstetrics union in the U.S., the positive impact of training during pregnancy can be justified by the fact that having knowledge is necessary for entering an unfamiliar, fearful environment. Accordingly, training the mothers provides them with positive attitude towards pregnancy and increases their self-confidence.¹⁷ A previous study revealed the positive effect of psychological training methods on reduction of anxiety during pregnancy.¹⁸ It has been reported that the mothers undergoing these trainings learn problem solving skills and consider pregnancy as a natural phenomenon. Healthy social relationships also increase in these classes.¹⁹ The mothers taking part in educational interventions show more maternal behaviors which can in turn affect their health. Additionally, maternal-fetal attachment can be a predictor of the primary relationship between mother and infant. Thus, increase of attachment during pregnancy results in more self-confidence for playing the maternal role and causes the mother to show more compatibility with postpartum conditions and give positive responses to infant's behaviors.²⁰ Using non-pharmacological methods is one of the best methods of coping with stress, creating tranquility, and increasing mother's attachment. Studies have shown that relaxation leads to tranquility and reduction of anxiety. Correct performance of relaxation improves blood circulation and decreases stress, anxiety, and depression by creating a positive attitude and improving the brain function. Relaxation also increases the individuals' cellular energy and confidence by reducing their anxiety.²¹ Considering the great effects of education during pregnancy and the contradictory results obtained regarding the effect of relaxation training on anxiety during pregnancy, the present study aims to assess the effect of relaxation training on the level of anxiety and maternal-fetal attachment in the primiparous women who had used IVF to get pregnant.

MATERIALS AND METHODS

The present semi-experimental clinical trial was conducted on 80 primiparous women referring to infertility clinics of Shiraz who had used IVF to get pregnant. At first, a 70-subject sample size (35 in each group) was determined for the study. Yet, considering the probable loss rate, the sample size was

increased to 40 subjects in each group. The inclusion criteria of the study were being primiparous, having used IVF to get pregnant, being 18–40 years old, having at least middle school degree, gestational age of 32–35 weeks, not having obstetric problems, and singleton pregnancy. On the other hand, the exclusion criteria of the study were suffering from cardiovascular diseases or serious psychological disorders and having pregnancy complications such as bleeding, diabetes, hypertension, and preterm labor.

The study data were collected using an interview form (including individual and pregnancy information), Spielberger's state-trait anxiety scale, Cranley's questionnaire (for assessing maternal-fetal attachment), and Avant's questionnaire (for assessing maternal-infantile attachment). Spielberger's scale consists of two sections, namely state and trait anxiety. The maximum and minimum scores of this scale are 80 and 20, respectively. Spielberger's scale has been widely used in Western research articles.^{22,23}

Cranley's maternal-fetal attachment questionnaire was first used by Cranley and its reliability and validity were determined.²⁴ After delivery, maternal-infantile attachment behaviors were assessed using Avant's checklist. This checklist was first used by Avant and its reliability and validity were confirmed.²⁵

In the current study, the participants were selected through convenience sampling and were randomly divided into an intervention ($n = 40$) and a control group ($n = 40$). The study is conducted according to Helsinki ethical principles. All In doing so, the first participant was allocated to the relaxation group, the second one was allocated to the control group, and the process continued. The researcher continuously referred to the selected treatment centers on the weekdays and asked the participants to fill out Spielberger's anxiety scale. In case the participants had low or average anxiety levels and signed written informed consents for taking part in the study, they were enrolled into the research. Before the intervention, all the participants were required to complete Cranley's maternal-fetal attachment questionnaire. Then, the mothers in the intervention group were divided into two 20-subject groups and participated in four 90-min educational classes weekly held on Saturdays. It should be mentioned that the intervention group received relaxation training in addition to the routine pregnancy care. The intervention was performed as follows (Table 1):

At the end of all the educational sessions, Benson's relaxation technique was performed. This technique is quite easy and safe and the mothers were reassured regarding its safety by a gynecologist. This method is very easy to learn and does not take more than 10–20 min to perform (from the tip of toes to forehead). Benson's relaxation technique can be done while lying down or sitting on the chair. In order to make sure that women did the relaxation technique

Table 1. The sessions of intervention.

First session	Anatomical, physiological, and hormonal changes during pregnancy, sperm and ovum, stages of IVF, physiology of fetal development, signs of anxiety, effects of anxiety on pregnancy, strategies for more adaptation with changes in pregnancy including appropriate nutrition, individual health, physical health, mental health, identification of methods for improving compatibility with changes in pregnancy such as relaxation, and effects of relaxation.
Second session	Development of fetus in different months of pregnancy, nutrition during pregnancy, effect of healthy nutrition and pregnancy care on maternal and fetal health, effect of relaxation on reduction of anxiety, and effect of relaxation on mother and fetus' physical and mental health.
Third session	Signs of risk in pregnancy and ways to cope with them, and effect of relaxation on improvement of mother's sleeping and nutrition, maternal-fetal attachment, and fetal growth.
Fourth session	Effect of relaxation on delivery process, necessity to perform relaxation during delivery, effect of relaxation on improvement after delivery, breastfeeding, and reduction of postpartum depression.

at home, in addition to an educational CD, they were provided with a checklist to record their daily exercises. The mothers were required to perform the relaxation technique at least once a day. At the beginning of each educational session, the checklists were collected, the educational content of the previous session was reviewed, and the participants' questions were answered. The most important practical trainings included staying in a calm environment, mental preparation, passive attitude, staying in a comfortable status, relaxation of all body muscles from the sole of the foot towards the upper extremities (for 10–20 min), and slowly getting out of relaxation.

The control group only received the routine care. It should be noted that the researcher carried out all the trainings. At the end of the intervention, both groups completed Spielberger's anxiety scale and Cranley's maternal-fetal attachment questionnaire again. After the delivery, the mothers' attachment behaviors were assessed during the first breastfeeding using maternal-infantile attachment checklist.

All the statistical analyses were performed using the SPSS statistical software (v. 18). In order to express the research participants' characteristics, descriptive statistics, including frequency, mean, and standard deviation (SD), were used. In addition, Chi-square was used to investigate the two groups' homogeneity regarding the qualitative variables. Besides, independent t-test was used to compare the two groups, while paired t-test was utilized to compare each group before and after the intervention. $P < 0.05$ was considered as statistically significant.

RESULTS

The study results revealed no significant difference between the two groups regarding age, level of education,

length of marriage, income level, satisfaction with one's husband, fetus sex, and housing status (Table 2).

The results of independent t-test showed no significant difference between the two groups regarding anxiety score before the intervention ($P = 0.903$). However, a statistically significant difference was observed in this regard after the intervention ($P < 0.001$). Besides, the results of paired t-test indicated that the anxiety score had significantly decreased in the intervention group ($P < 0.001$), but had significantly increased in the control group ($P = 0.033$). Thus, relaxation training was effective in reduction of anxiety score after the intervention (Table 3).

Based on the results, the means of both state and trait anxiety reduced in the intervention group and increased in the control group after the intervention. The results of independent t-test revealed no significant difference between the two groups regarding state and trait anxiety before the intervention. However, a significant difference was observed in this regard after the intervention (Table 4).

The results of independent t-test demonstrated no significant difference between the two groups concerning attachment scores before the intervention ($P = 0.918$). Nevertheless, this difference was statistically significant after the intervention ($P < 0.001$). Thus, relaxation training was effective in increasing maternal attachment in the intervention group compared to the control group. According to the results of paired t-test, attachment had significantly increased in the intervention group ($P < 0.001$), but no significant difference was observed in the control group in this regard after the intervention ($P = 0.374$) (Table 5).

Comparison of the mothers' attachment behaviors during the first breastfeeding showed that the mean of

Table 2. Distribution of primiparous women in the two groups based on demographic indexes.

Index	Intervention group Number (percent)	Control group Number (percent)	Statistics
Education level			
Middle school	7 (17.5)	13 (32.5)	Chi-square
High school and diploma	23 (57.5)	20 (50)	Df = 2
Academic	10 (25)	7 (17.5)	P = 0.110
Housing status			
Owner	11 (27.5)	18 (45.0)	Chi-square
Tenant	29 (72.5)	22 (55.0)	Df = 1
			P = 0.106
Satisfaction with one's husband			
Yes	37 (92.5)	34 (85)	Chi-square
No	3 (7.5)	6 (15)	Df = 1
			P = 0.292
Fetus sex			
Boy	19 (47.5)	18 (45)	Chi-square
Girl	15 (37.5)	15 (37.5)	Df = 1
Unknown	6 (15)	7 (17.5)	P = 0.784
Average income			
Below 5 million Rials	22 (55)	25 (62.5)	Chi-square
Above 5 million Rials	18 (45)	15 (37.5)	Df = 1
			P = 0.413
Occupation			
Employee	10 (25)	9 (22.5)	Chi-square
Homemaker	30 (75)	31 (77.5)	Df = 1
			P = 0.794
Age (mean $\pm$ SD)	29.0 $\pm$ 2.4	28.7 $\pm$ 2.7	Independent t-test
			Df = 78
			P = 0.672

attachment behaviors was higher in the intervention group compared to the control group. Various dimensions of behaviors were also compared between the two groups. Accordingly, the highest and lowest mean scores in both study groups were related to proximity and care behaviors, respectively. The results of independent t-test indicated a significant difference between the two groups regarding emotional behaviors (first dimension, $P < 0.001$), proximity behaviors (second dimension, $P < 0.001$), and mother's attention to infant (fourth dimension, $P < 0.001$). Also, a significant difference was observed between the two groups concerning the total mean score of attachment behaviors ($P < 0.001$). The maternal-infantile attachment score showed a

considerable increase in the intervention group in comparison to the control group (Table 6).

DISCUSSION

Since pregnancy is an important and sensitive period for women, training the pregnant women can be of great help.²⁶ Anxiety has major effects on women's health during pregnancy which must be decreased through different psychological and social skills.²⁷ The more these skills are easy, accessible, and without complications, the more they will be used by individuals to meet their needs and solve their problems.²⁸ According to studies, relaxation is the most effective treatment method for

Table 3. Comparison of mean of anxiety in primiparous women in the two groups.

Group	Variable			Results (paired t-test)
	Before the intervention (Mean $\pm$ SD)	After the intervention (Mean $\pm$ SD)	Difference between before and after the intervention (Mean $\pm$ SD)	
Intervention	47.8 $\pm$ 5.2	41.4 $\pm$ 4.1	-6.4 $\pm$ 6.2	T-test: $t = 6.5$ $P < 0.001$ $df = 39$
Control	47.7 $\pm$ 4.9	51.0 $\pm$ 10.4	3.2 $\pm$ 9.4	T-test: $t = -2.2$ $P = 0.033$ $df = 39$
Results (independent t-test)	T-test: $t = 0.122$ $P < 0.903$ $df = 78$	T-test: $t = -5$ $P < 0.001$ $df = 78$	T-test: $t = -5.4$ $P < 0.001$ $df = 78$	

psychosomatic disorders.²⁹ Relaxation techniques have been proved to be effective in treatment of delivery pain, anxiety, and insomnia and cause the individuals to control their emotions and behaviors.²⁹ Response to relaxation is a physiological phenomenon which is activated by the parasympathetic nervous system and results in reduction of anxiety by facilitating the secretion of endorphins. After muscle relaxation and alteration of sympathetic to parasympathetic dominance, individuals feel relaxed. In this way, people's attention is distracted from external issues and is rather focused on the physical activity. Therefore, individuals will spend more time in tranquility and will experience less irritability and anxiety in future.³⁰

The findings of the present study showed that relaxation training reduced the anxiety level in the intervention group. The reduction was observed in both state and trait anxiety. This might be due to cognitive restructuring.

In this way, individuals develop correct and logical knowledge and will have a safer interpretation of the situation. This change in the level of anxiety and its dimensions is due to the effect of training in the pregnancy period, because training can play a critical role in reduction of psychological disorders as well as anxiety and improvement of physical and mental health. To date, nursing staff employ various interventions to control anxiety which can also be of great help in prevention of anxiety disorders.³¹ Moreover, educational interventions during pregnancy, particularly in high-risk populations, can be highly effective in prevention of mortality and reduction of prenatal complications, depression, anxiety, and their subsequences. Overall, training during pregnancy plays a key role in reduction of mental disorders and anxiety and improvement of physical as well as mental health.³¹ The results of the present study were in

Table 4. Comparison of mean of anxiety dimensions in primiparous women in the two groups.

Variable		Group		Results (independent t-test)
		Intervention (Mean $\pm$ SD)	Control (Mean $\pm$ SD)	
State anxiety	Before the intervention	24.6 $\pm$ 2.9	24.7 $\pm$ 2.8	T-test: $t = -0.286$ $P = 0.776$ $df = 78$
	After the intervention	21.9 $\pm$ 2.7	25.0 $\pm$ 3.9	T-test: $t = -3.1$ $P = 0.001$ $df = 78$
Trait anxiety	Before the intervention	23.2 $\pm$ 2.9	22.9 $\pm$ 3.0	T-test: $t = -4.197$ $P = 0.628$ $df = 82$
	After the intervention	19.5 $\pm$ 2.9	25.9 $\pm$ 7.5	T-test: $t = -5.021$ $P = 0.001$ $df = 78$

Table 5. Comparison of mean of attachment in primiparous women in the two groups.

Group	Before the intervention (Mean $\pm$ SD)	After the intervention (Mean $\pm$ SD)	Difference between before and after the intervention (Mean $\pm$ SD)	Results (paired t-test)
Intervention	61.1 $\pm$ 4.4	67.0 $\pm$ 2.9	5.8 $\pm$ 5.2	T-test: $t = -7.0$ $P < 0.001$ $df = 39$
Control	61.0 $\pm$ 4.1	62.0 $\pm$ 5.5	0.9 $\pm$ 6.6	T-test: $t = -0.8$ $P = 0.374$ $df = 39$
Results (independent t-test)	T-test: $t = 0.104$ $P = 0.918$ $df = 78$	T-test: $t = 5.0$ $P < 0.001$ $df = 78$	T-test: $t = -3.6$ $P = 0.001$ $df = 78$	

agreement with those of the studies conducted by Teixeira, Chang, and Davis in which the performed interventions decreased the mothers' anxiety level.^{32–34}

The findings of the current study demonstrated an increase in attachment after relaxation training in the intervention group, while no significant increase was observed in control group's attachment. Educational interventions during pregnancy can train the pregnant women regarding their necessary issues and, consequently, eliminate their problems during the pregnancy period. In fact, increasing the mothers' knowledge and readiness during pregnancy helps them to pass this period more desirably and with fewer complications. Therefore, pregnancy provides an appropriate opportunity to increase women's knowledge about the advantages of training and having a healthy lifestyle.³⁵ In general, anxiety control during pregnancy is necessary for maintaining the mother's mental health and is directly associated with attachment. Mothers with high attachment consider themselves to be quite capable of keeping the fetus healthy. Those with low attachment, on the other hand, are faced with lots of problems, are more concerned with others' external control, and are not sure about their ability to maintain the fetus' health.³⁶ Increase in attachment

might be due to the fact that educational interventions can play a role in reduction of mental disorders and anxiety and improvement of physical and mental health.³¹

According to the results of the present study, maternal-infantile attachment was significantly higher in the intervention group compared to the control group. This might be due to the educational content, more interaction, stronger imagination, and more relationship with the fetus in the intervention group. The mothers receiving these interventions show more maternal behaviors which can in turn affect their mental health and decrease their anxiety level. Also, increase of attachment during pregnancy can provide the mother with more self-confidence and compatibility with postpartum conditions eventually leading her to respond positively to the infant's behaviors.²⁰ Since anxious personality can be eliminated through education, training the infertile women can help them a great deal to have more control on the stressful events of their life. Thus, programs such as relaxation techniques increase of self-confidence, gaining the necessary training, and utilization of supportive systems should be conducted for infertile women. Such trainings can effectively reduce tension resulting from infertility and its treatment, increase attachment and

Table 6. Mean and SD of various dimensions and total maternal-infantile attachment during breastfeeding in the two groups.

Various dimensions of maternal-infantile attachment behaviors	Intervention	Control	Independent t-test
	Mean $\pm$ SD	Mean $\pm$ SD	
Emotional behaviors (first dimension)	28.6 $\pm$ 7.1	16.7 $\pm$ 8.6	$P < 0.001$
Proximity behaviors (second dimension)	45.0 $\pm$ 0.0	43.5 $\pm$ 2.4	$P < 0.001$
Care behaviors (third dimension)	6.2 $\pm$ 3.4	5.3 $\pm$ 3.3	$P = 0.273$
Mother's attention to infant (fourth dimension)	12.1 $\pm$ 2.8	8.3 $\pm$ 3.7	$P < 0.001$
Total attachment behaviors	92.0 $\pm$ 10.8	74.0 $\pm$ 13.8	$P < 0.001$

trust, and finally improve these women's health (see Tables 5 and 6).

CONCLUSIONS

The study results showed that relaxation training was effective in reduction of anxiety and increase of maternal-fetal attachment in the women who had used IVF to get pregnant. Although there are other methods for reducing anxiety, relaxing can be used for IVF women due to its ease of use and its wide acceptance.

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